

```
#include <stdio.h>
#include <stdlib.h>

#define MAX 5

int stack[MAX];
int top = -1;

/* Function Prototypes */

void push();
void pop();
void peek();
void traverse();

/* Main Function */

int main()
{
    int choice;

    while(1)
    {
        printf("\n--- Stack Menu ---\n");
        printf("1. Push\n");
        printf("2. Pop\n");
        printf("3. Peek\n");
        printf("4. Traverse\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d",&choice);

        switch(choice)
        {
            case 1: push();
                    break;

            case 2: pop();
                    break;

            case 3: peek();
                    break;
```

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        case 4: traverse();
                break;

        case 5: exit(0);

        default: printf("Invalid choice\n");
    }
}

return 0;
}

/* Push Operation */

void push()
{
    int value;

    if(top == MAX-1)
    {
        printf("Stack Overflow \n");
        return;
    }

    printf("Enter value to push: ");
    scanf("%d",&value);

    top = top + 1;
    stack[top] = value;

    printf("Element inserted\n");
}

/* Pop Operation */

void pop()
{
    if(top == -1)
    {
        printf("Stack Underflow\n");
        return;
    }

    printf("Deleted element is %d\n",stack[top]);
}

```

```
    top = top - 1;  
}
```

```
/* Peek Operation */
```

```
void peek()  
{  
    if(top == -1)  
    {  
        printf("Stack is empty\n");  
        return;  
    }  
  
    printf("Top element is %d\n",stack[top]);  
}
```

```
/* Traverse Stack */
```

```
void traverse()  
{  
    int i;  
  
    if(top == -1)  
    {  
        printf("Stack is empty\n");  
        return;  
    }  
  
    printf("Stack elements are:\n");  
  
    for(i = top; i >= 0; i--)  
    {  
        printf("%d\n",stack[i]);  
    }  
}
```

Compile Result

--- Stack Menu ---

1. Push
2. Pop
3. Peek
4. Traverse
5. Exit

Enter your choice: 1

Enter value to push: 2

Element inserted

--- Stack Menu ---

1. Push
2. Pop
3. Peek
4. Traverse
5. Exit

Enter your choice: 2

Deleted element is 2

--- Stack Menu ---

1. Push
2. Pop



GIF



1

2

3

4

5

6

7

8

9

0

Compile Result

--- Stack Menu ---

1. Push
2. Pop
3. Peek
4. Traverse
5. Exit

Enter your choice: 1

Enter value to push: 2

Element inserted

--- Stack Menu ---

1. Push
2. Pop
3. Peek
4. Traverse
5. Exit

Enter your choice: 3

Top element is 2

--- Stack Menu ---

1. Push
2. Pop



GIF



1

2

3

4

5

6

7

8

9

0